A-A-50220
20 January 1984

COMMERCIAL ITEM DESCRIPTION

LATHE, AXLE, JOURNAL-TRUING

RAILWAY, DUAL OR SINGLE END DRIVE

The General Services Administration has authorized the use of this commercial item description

This commercial item description covers the requirements for a manually operated journal-truing axle lathe for remachining bare axles and the journals of mounted wheel sets.

Salient characteristics:

Capable of handling axle lengths between 66 inches and 104 inches.
Capable of handling wheel tread diameters 42 inches.
Capable of turning axles between 3-3/8 inches to 12-1/2 inches diameter.
Capable of burnishing axles between 4 inches to 12-1/2 inch diameter.
Capable of accepting axles with or without mounted wheel.
Infinitely variable feedrate.
Spindle speeds shall range between 125 - 250 rpm.
Self-aligning headstocks.
Manual or automatic cross slide-positioning.
Main drive motors shall be two speed 440V AC.
Centers at 60^{o}.
Tool holders with a minimum of twelve carbide tool inserts provided.
Tool holder swivels.
Burnishing tools.
Drive ring sets for minimum of three different axle sizes.
Painting and finishing of the machine shall-be the manufacturer’s standard commercial practice.
Permanent metal identification plate mounted in a conspicuous place and showing manufacturer's name, address, contract number, model, serial number, date of manufacture, and power input characteristics and rating.

Design and construction. All components shall be designed to form an integral part of a compact unit conforming to the requirements listed herein. The unit shall be designed with ready access to components for adjustments, replacement, or repair, and to facilitate set-up, operation and maintenance without the need for special tools. Special tools that are required shall be provided by the contractor. The machine shall be complete so that when installed and connected to the specific power source, it can be used for any operation for which it was designed. Construction shall be such that the unit will retain its performance and accuracy, and function properly when operated at its maximum functional capacity.

FSC 4940

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Materials and workmanship. Materials shall meet ASTM specifications where applicable for the purpose intended. The machine shall be fabricated from compatible materials, inherently corrosion resistant or treated to be protected against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed. Dissimilar metals as defined in MIL-STD-889 shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Workmanship in construction and finish shall be of high quality and consistent with industry standards. The machine shall be constructed of parts which are sound in structure and free from any repair. All parts shall be new and unused. Castings, forgings molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled, except where a sharp edge is required by design.

Essential Safety Features. The machine shall be fitted with suitable devices and covering or guards to provide protection from electrical shock and moving parts such as gears, belts, pulleys, and other parts which may present a hazard. All guards shall be designed to provide ready access to the guarded parts. All controls shall be within convenient reach and vision of operator in his normal operating position. The lathe shall meet all applicable OSHA requirements. The machine shall be provided with a prominently located warning sign intended for operator protection. It shall alert the operator to the noise and eye hazard which will require safety glasses and ear plugs.

Lubrication requirements. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with a means for lubrication. All oil holes, grease fittings, and filler caps shall be easily accessible. A lubrication chart or. plate shall be securely attached to the unit. The chart shall be placed in a transparent plastic folder or laminated between clear plastic sheets and permanently sealed and provided with suitable means for mounting on the unit. The chart or plate shall provide the following information:

POINTS OF LUBRICATION
SERVICE INTERVAL
TYPE OF LUBRICATION
VISCOSITY

Standard commercial product. The above salient characteristics are set forth as minimum requirements. The lathe shall be in accordance with these minimum requirements and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this description but which are a part of the manufacturer's standard commercial product, shall be included in the lathe being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures, and represents the latest production model. It shall have been in production, marketed, and in use for a minimum of one year. The introduction of normal product improvements is acceptable. The bidder shall submit proof of this offering as a commercial product with his bid. The contracting officer will determine the acceptability of this proof . Failure of the bidder to furnish proof of his offering as his standard commercial product shall be cause for rejection of his bid.

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Certification. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product sold in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Acceptance. Acceptance of each lathe shall be based on visual inspection and demonstration of its performance capabilities. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements. The Government shall completely inspect the lathe for design, construction, materials, components, electrical equipment and workmanship to determine compliance with these requirements. The lathe shall be given at least a 1 hour continuous run test combining all operations and all speed ranges up to maximum speed permissible. Any deviation from these requirements shall be cause for rejection.

Packaging and marking. Military packaging shall be specified when shipments are to overseas Department of Defense (DOD) consignees. Commercial packaging may be specified for shipments to DOD consignees within the Continental United States and to non-DOD consignees worldwide. When a contractors norma1 packaging will satisfy the requirements for military packaging, it may be used.

Military packaging. Each lathe shall be preserved and packed to protect against physical damage and deterioration in a logistic environment characterized by the following:

1. Multiple handling during transportation and intransit storage.
2. Shock, vibration, and static loading forces during worldwide shipment by all transport modes.
3. Uncontrolled warehouse storage for extended periods.
4. Weather exposure during shipment and intransit transfers.
5. Stacking loads during shipment and storage.

Commercial. Each lathe shall be packaged in accordance with the contractor’s normal method for retail distribution. Shipping containers shall comply with Uniform Freight Classification or National Motor Freight Classification rules.

Marking. Unit, intermediate (when provided), and exterior packs shall be marked with the item nomenclature, stock or part number, quantity per pack, and contract number as a minimum; additional marking shall be as specified in the contract or order. Exterior packs shall have in addition the complete consignee's address. Markings shall be applied in legible manner, in a contrasting color, of an appropriate size and waterproof materials.

Custodian
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PREPARING ACTIVITY
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Project 4940-A454

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